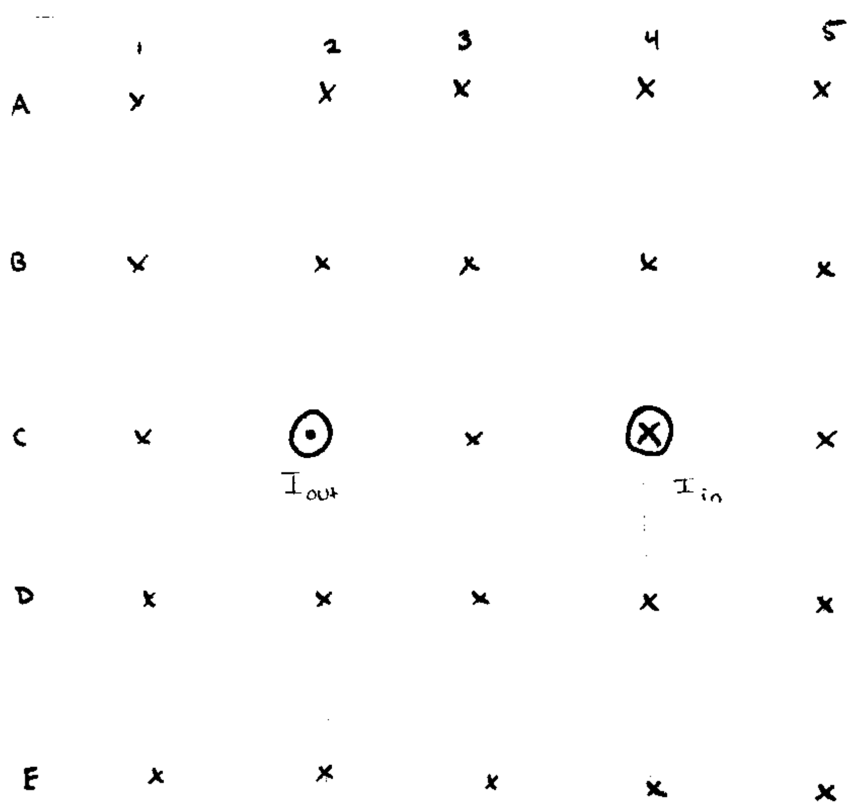
Name: ____

Physics 112 - Exam 2 - W26

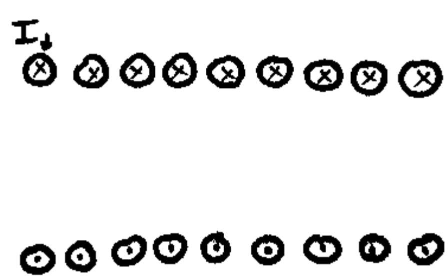
Problem 1

Create a diagram of the magnetic field around the following wires. Calculate the field at each of the points and finally try to connect the grid together to show field lines.



Problem 2

Sketch the magnetic field for a solenoid. Try to be accurate.



Problem 3

In a modern U.S. power plant, electricity is generated by rotating a coil inside a magnetic field. The generators rotor consists of a circular coil made of copper wire with 12,000 turns. The coil rotates at a constant speed of 3600 revolutions per minute (rpm) in a uniform magnetic field of strength 0.15 *Tesla*. The generator is designed to produce a peak electromotive force (EMF) of 18 kV.

a) Find the required area A of the coil to create this EMF.

b) Assuming the coil is circular, express its area as $A = \pi R^2$. Calculate the radius R of the coil required to achieve the target peak voltage.

c) After generation, the voltage is stepped up for transmission over long distances. If a transformer is used to raise the voltage from 18 kV to 115 kV, what is the ratio of secondary turns to primary turns in the transformer.

Problem 4

A mass spectrometer is used to analyze a gas sample suspected to contain one of several ionized species. In this instrument, ions first pass through a velocity selector before entering the mass analyzer. The velocity selector has a uniform electric field $E = 5000\,\mathrm{V/m}$ and a perpendicular magnetic field $B = 0.1\,\mathrm{T}$ are set up. After selection, the ions enter a region with a uniform magnetic field $B = 0.2\,\mathrm{T}$ and follow a circular path of radius $r = 0.085\,\mathrm{m}$. Assume the ions are singly charged (with charge $q = 1.6 \times 10^{-19}\,\mathrm{C}$).

a) Calculate the speed v of the ions that pass through the velocity selector.

b) When an ion of mass m and charge q is shot into the magnetic field B', the particle travels in a circular path. Use the Lorrentz force to calculate the numerical value of $\frac{m}{q}$.

c) The sample is known to possibly contain one of the following ions:

- N_2^+ (molecular mass $\approx 28\,\mathrm{u}$),
- CO^+ (molecular mass $\approx 28\,\mathrm{u}$), or
- O_2^+ (molecular mass $\approx 32\,\mathrm{u}$).

Given that $1\,\mathrm{u} = 1.66 \times 10^{-27}\,\mathrm{kg}$, use your result from part (b) to determine which ion is most consistent with the measured $\frac{m}{q}$ value.